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Effects of micro-aeration on microbial niches and antimicrobial resistances in blackwater anaerobic digesters

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ABSTRACT

Anaerobic digestion (AD) of source-diverted blackwater (toilet flush) at ambient room temperature presents challenges for fast hydrolysis of particulate matters. This study investigated the effect of different micro-aeration dosages for blackwater AD. Sequencing batch reactors were operated at ambient room temperature (22 ± 1°C) with micro-aeration (0, 5, 10, 50, and 150 mg O_2 g^{-1} COD_{feed} per cycle) and gradually reduced hydraulic retention times from 5 d to 2 d. The methanogenesis efficiencies were greater at low oxygen dosages (*i.e.*, 0, 5, 10) while the volatile fatty acids (VFAs) accumulated more at high oxygen dosages (*i.e.*, 50, 150). Microbial communities were significantly different under different oxygen dosages ($p<0.05$), with segregation of microbial ecological niches in low and high oxygen dosage communities. The low-oxygen-dosage niche (0, 5, and 10 mg g^{-1} COD_{feed}) was inhabited by fermenting and syntrophic bacteria (*e.g., Cytophaga, Syntrophomonas*) and methanogens (*e.g., Methanobacterium, Methanolinea, Methanosaeta*). The high-oxygen-dosage niche (50 and 150 mg g^{-1} COD_{feed}) had significantly ($p<0.05$) more facultative anaerobic bacteria (*Ignavibacteriales* and *Cloacamonales*), and aerobic bacteria (*Rhodocyclales*). Moreover, blackwater can be a source of antimicrobial resistance genes (ARGs), which are affected by different oxygen dosages. The ARG variation correlated with the microbial community composition ($p<0.05$). Low-oxygen-dosage communities contained a higher prevalence of mobile gene elements (*intI1* and *korB*) and *tetM, ermB, sul1, sul2*, and *blaCTX-M* than the high-oxygen-dosage communities, indicating that oxygen dosage influenced the prevalence of populations carrying ARGs. These findings suggest that application of micro-aeration to AD can be used to control ARG profiles.

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1. Introduction

Decentralized source-diverted municipal wastewater treatment provides opportunities for high-efficiency resource recovery, mainly methane production from organic waste (Florentino et al., 2019; Gao et al., 2019b; Yu et al., 2020). Toilet-flushed blackwater contains high concentrations of organic substrates that can be converted into biomethane using anaerobic digestion (AD) processes (Florentino et al., 2019; Gao et al., 2019a; Gao et al., 2019b; Xu et al., 2019; Zhang et al., 2020). Conventionally, AD is performed at thermophilic (~55 °C) or mesophilic (~35 °C) temperatures to maintain high microbial activity and treatment efficiencies (Zhang et al., 2020). In temperate and cold regions, the heating requirement for thermophilic and mesophilic AD reactors requires great energy consumption. Compared to conventional thermophilic and mesophilic conditions, AD operation at ambient room temperature (~20 °C) reduces the capital cost and energy consumption for reactor heating (Guo et al., 2020; Keating et al., 2018). However, the hydrolysis rate is often limited at ambient room temperature or lower temperatures, which may result in reduced net energy yields (Rose et al., 2015; Zhang et al., 2013).

To improve hydrolysis efficiency, pre-treatment techniques and co-digestion have been previously investigated (Zhang et al., 2013). One of the low-cost strategies is to use single-reactor configuration with micro-aeration to enhance *in-situ* hydrolysis in AD reactors at low temperatures. Although oxygen is recognized as an inhibitory agent for strict anaerobes, especially methanogens in AD, at proper dosages it can be quickly consumed by facultative anaerobes and facilitate the hydrolysis of particulate matters and large molecules (Botheju and Bakke, 2011; Jagadabhi et al., 2010; Johansen and Bakke, 2006; Montalvo et al., 2016; Tsapekos et al., 2017; Zhu et al., 2009). Few studies have investigated the microbial community dy-

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namics caused by micro-aeration in AD processes, either in batch experiments (Duarte et al., 2018; Fu et al., 2016) or in continuously operated reactors (Nguyen et al., 2019; Yu et al., 2020). Microbial community structure and functionality are altered by changing environmental factors, which form ecological niches under different oxygen concentrations. Micro-aeration can lead to changes in microbial community composition, predominant species, and stable microbial niches under long-term operation. Yu et al. (2020) investigated the relationship between oxygen input dosage, hydrolysis efficiency, and the methane production rate of blackwater AD treatment. The oxygen input can promote hydrolysis of particulate organic matters and improve the production of volatile fatty acids (VFAs) while high oxygen dosages may inhibit strict anaerobic bacteria and methanogens and lead to inhibited methane production. Given the different end products (methane or VFAs) controlled by the micro-aeration strategy, it remains unclear how the microbial niches developed gradually in the micro-aeration facilitated AD, and the key marker microorganisms are yet to be explored.

In addition to the formation of microbial niches, micro-aeration may affect the prevalence of human-associated pollutants such as antimicrobial resistance genes (ARGs). Of particular concern, wastewater treatment systems are recognized as reservoirs of ARGs (Christgen et al., 2015; Du et al., 2015; Yin et al., 2019; Yuan et al., 2016). While a few previous ARG studies have focused on rural wastewater treatment systems (Chen and Zhang, 2013; Zhou et al., 2018), most studies have focused on centralized municipal wastewater treatment systems; studies on decentralized systems are limited. Further, there have been no studies to date that have investigated ARG prevalence in microbial communities in ADs treating blackwater. Since blackwater contains complex organic substrates and human-associated ARGs, it continuously introduces ARGs to the AD microbial communities. Comparison of different wastewater types revealed that human feces contain higher ARGs than mixed-source municipal wastewater (Yee et al., 2019; Zhang et al., 2019; Zhou et al., 2018). AD treatment has been reported to effectively reduce ARGs in municipal wastewater or waste activated sludge (Wu et al., 2018; Xiao et al., 2020).

Previous studies comparing anaerobic and aerobic processes for wastewater treatment differ in their findings. Some studies showed that aerobic digestion of waste activated sludge resulted in higher ARGs than AD (Burch et al., 2017; Du et al., 2015), while other studies showed that AD sludge carried more ARGs than aerobic sludge (Christgen et al., 2015; Yang et al., 2014; Yin et al., 2019; Yuan et al., 2016). No studies have investigated the effects of different oxygen levels on ARGs in AD systems. There is a need to better understand dynamics of populations carrying ARGs in the micro-aeration conditions and how they are affected by oxygen dosages in order to provide insights into the association between microbial niches and ARG prevalence.

The aims of this study were to (i) investigate the microbial niches formed by micro-aeration during blackwater anaerobic digestion at ambient room temperature and reveal the marker microorganisms in different niches and (ii) examine the impacts of oxygen dosages on the prevalence of ARGs and the variation of microbial communities.

2. Material and Methods

2.1. Sequencing batch reactors

Five sequencing batch reactors (SBRs) with volumes of 1 L and working volumes of 0.71 L were fed with blackwater (1000 mg COD/L) collected on campus (University of Alberta, Edmonton, Canada) and stored at 4°C prior to processing. The soluble COD varies from 30-50% during the sample storage time,which was determined at the start of each cycle and used for all calculations. The hydraulic retention time (HRT) was sequentially reduced to allow adaptation and reach a comparable HRT with previous studies at 35 °C (Gao et al., 2019a; Gao et al., 2020). HRT was stepwise decreased when stable hydrolysis and methanogenesis were observed. Stage 1 (HRT 5 d), Stage 2 (HRT 4 d), Stage 3 (HRT 3 d) and Stage 4 (HRT 2 d) were operated for 35, 20, 30 and 18 days, respectively. The sludge retention time was not controlled. Each cycle consisted of filling (5 min), reaction (time according to HRT), settling (30 min), and discharging (5 min). The reactors were inoculated with 0.5 g volatile solids anaerobic sludge from an anaerobic digester and 0.5 g volatile solids aerobic activated sludge from an aeration tank in a full-scale wastewater treatment plant (Alberta, Canada) to include both anaerobic and aerobic microorganisms.

At the beginning of each cycle, 0.5 L of feed blackwater was supplied to each reactor (*i.e.*, volume exchange ratio of 70%). Micro-aeration conditions were achieved with injection of air (21% oxygen) into the sealed reactors at the beginning of each cycle. The volumes of air injected to create oxygen dosages of 0, 5, 10, 50, and 150 mg g^{-1} COD_{feed} for the five SBRs were calculated as below:

$$V_{air\ injected} = \frac{Dosage_{oxygen}/MW_{oxygen} \cdot R \cdot T \cdot COD_{feed}}{P \cdot Percentage_{oxygen\ in\ air}} \quad (1)$$

Where MW_{oxygen} is the molecular weight of oxygen (32 g $mole^{-1}$), R is 0.0821 L•atm mol^{-1}•K^{-1}, T is 295.15 K (22 °C), COD_{feed} is substrate fed per cycle (0.5 g COD), P is the air pressure (1 atm), $Percentage_{oxygen\ in\ air}$ is 21%.

The reactors were placed on a shaker at 120 rpm at ambient room temperature (22 ± 1 °C) protected from light. At the end of the cycle, reactors were settled for 30 min and the supernatant was discharged.

2.2. Analytical methods

The effluent was measured for pH using a Symphony pH probe (VWR, Radnor, USA) and filtered through a 0.2 μm membrane filter for VFA (*i.e.*, acetate, propionate, butyrate) concentrations using an ionic chromatography system (DIONEX ICS-2100, Thermo Fisher Scientific, Waltham MA, USA) equipped with a conductivity detector and IonPac AS18 Analytical Column. Chemical oxygen demand (COD), soluble COD (sCOD, filtered at 0.45 μm), total suspended solids (TSS), and volatile solids concentrations were measured according to the Standards Methods 5220D, 5220D, 2540D and 2540E, respectively (APHA and AWWA, 2012). The produced gas volume and composition were measured at the end of each cycle. Biogas composition was analyzed with a 7890B gas chromatograph (Agilent Technologies, Santa Clara USA) equipped with two columns (Molsieve 5A 2.44 m 2 mm for CH_4 and Hayesep N 1.83 m 2 mm for N_2, O_2 and CO_2) and a thermal conductivity detector (TCD).

2.3. Methanogenesis and hydrolysis efficiency

The methanogenesis efficiency was calculated using Equation 2, and the hydrolysis efficiency was calculated using Equation 3, where CH_4 is in COD units.

$$Methanogenesis\ efficiency\ (\%) = \frac{CH_4}{TCOD_{influent}} \quad (2)$$

$$Hydrolysis\ efficiency\ (\%) = \frac{SCOD_{effluent} + CH_4 - SCOD_{influent}}{TCOD_{influent} - SCOD_{influent}} \quad (3)$$

2.4. DNA extraction, PCR and sequencing

At the end of each operation stage, 2-ml sludge samples were taken from each reactor for DNA extraction. The DNA extraction was performed using DNeasy PowerSoil Kit (QIAGEN, Toronto,

Canada) following the manufacturer's protocol. The quality and quantity of extracted DNA were examined using agarose gel and NanoDrop One (ThermoFisher, USA). The V4 hypervariable regions of the microbial 16S rRNA genes were amplified using the modified universal primers 515F (5'-GTG CCA GCM GCC GCG G-3') and 806R (5'-GGA CTA CHV GGG TWT CTA AT-3') that can efficiently cover both bacteria (95% coverage of known 16S rRNA gene sequences available on the Silva database) and archaea (92% coverage) at same time (David G. et al., 2020), following the conditions described by Guo et al. (2020). Tag sequences were added to forward (5' - ACA CTG ACG ACA TGG TTC TAC A-3') and reverse (5'- TAC GGT AGC AGA GAC TTG GTC T-3') primers for downstream barcoding. The amplicons were barcoded and sequenced at the Génome Québec Innovation Centre (Montréal, QC, Canada) using the Illumina Miseq PE250 platform. The raw sequence data files are deposited at NCBI GenBank project (PRJNA701935).

2.5. Data analysis

Raw sequences were analyzed using the Qiime2 DADA2 pipeline (Callahan et al., 2016; Caporaso et al., 2010) with 99% similarity with reference to the SILVA database (version 132). Alpha and Beta diversities, Principal Coordinates Analysis (PCoA) of Bray-Curtis distance, Canonical Correspondence Analysis (CCA), ANOVA of CCA were performed using the "vegan" package (Jari Oksanen et al., 2017) in RStudio version 3.4.1. Linear Discriminant Analysis Effect Size (LEfSe) analysis was performed to rank the differentially distributed taxa, which are significantly different between groups (Kruskall-Wallis test) and consistent within groups (pairwise Wilcoxon test) (Segata et al., 2011). Analysis of similarities (ANOSIM) tests were used to test community differences among oxygen dosages and among different HRTs. The Mantel test was performed to examine the correlation (Pearson's product-moment correlation) between microbial community and ARGs' abundances. Two-way ANOVA test was performed between the ARG relative abundances and two factors, HRT and oxygen dosage.

2.6. Relative quantification by qPCR

A set of ARGs commonly evaluated in sewage (Pallares-Vega et al., 2019) were relatively quantified, including mobile gene elements class 1 integrase (*intI1*) and *korB* which is specific for IncP-1 plasmid RK2, and ARGs tetracyclines (*tetM*), macrolides (*ermB*), sulfonamides (*sul1* and *sul2*), extended-spectrum beta-lactamases (*blaCTX-M*), and fluoroquinolones (*qnrS*). The bacterial RNA polymerase beta subunit (*rpoB*) gene was quantified with qPCR and standard curves using the method described in Florentino et al. (2019). The relative abundance of ARGs with reference to bacterial *rpoB* was used to represent the prevalence of ARGs ($2^{C_{T\,rpoB}-C_{T_{target}}}$). Compared to 16S rRNA gene, *rpoB* has a key advantage of harbouring only a single copy in the genome of microbial cells (Dahllöf et al., 2000). The primers and conditions are summarized in Supplementary Table S1.

3. Results and Discussions

3.1. Reactor performances affected by micro-aeration

The reactor methanogenesis efficiencies, VFA concentrations, and organic matter hydrolysis efficiencies are shown in Fig. 1. From Stage 1 to 4, the hydrolysis efficiency was increased from 16.6-37.7% (anaerobic) to 41.3-53.1% with the oxygen dosage of 5 mg g^{-1} COD_{feed}. It reached the lowest level (17.9-26.5%) with the highest oxygen dosage (150 mg g^{-1} COD_{feed}), which can be attributed to inhibition effect of oxygen to hydrolyzing bacteria and the adverse feedback of downstream methanogenesis and acidogenesis

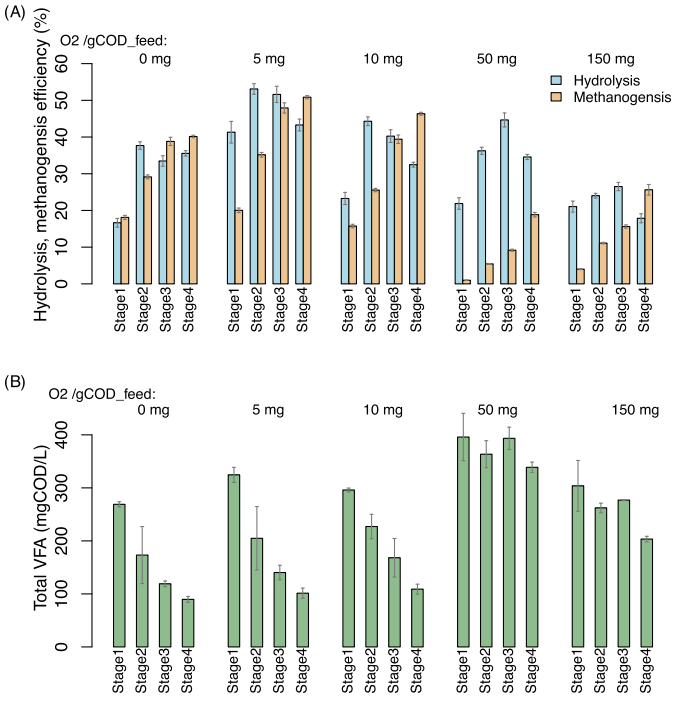


Fig. 1. Reactor performance with different oxygen dosages in the last cycle of each stage. From Stage 1 to 4, the HRT was sequentially reduced from 5 d to 4 d, 3 d, and 2 d after reaching stable performance. (A) Hydrolysis and methanogenesis efficiencies. (B) Total VFA concentrations. Error bar represents standard deviation of two measurements.

processes if inhibited by oxygen. The anaerobic and low-oxygen-dosage reactors (0, 5, and 10 mg g^{-1} COD_{feed}) showed higher methanogenesis efficiencies (18.1-40.1%, 20.0-50.9%, 15.7-46.3%, respectively) than the high-oxygen-dosage reactors of 50 and 150 mg g^{-1} COD_{feed} (1.0-18.9% and 4.0-25.6%, respectively). Under all HRT conditions, the highest methanogenesis efficiency was observed when the lowest oxygen dosage was applied (5 mg g^{-1} COD_{feed}) with oxidation-reduction potential (ORP) of -180 mV (Supplementary Fig. S1). Higher oxygen dosage led to higher ORP (-308, -180, -142, -109 and -69 mV for 0, 5, 10, 50 and 150 mg g^{-1} COD_{feed} respectively, Supplementary Fig. S1), which could affect the survival and activity of methanogens. Other studies have reported that ORP of -265 mV and -180 mV may improve methanogenesis when eliminating sulphide toxicity (Khanal Samir and Huang, 2003; Khanal and Huang, 2003; Nguyen et al., 2019), and intermittent micro-aeration at 50 mV higher than baseline ORP can be inhibitory for methane production (Khanal and Huang, 2003). The particulate COD was purposely retained in the reactor, and the COD reduction from influent to effluent showed similar rates for low-oxygen-dosage reactors (0, 5, and 10 mg g^{-1} COD_{feed}) (Supplementary Fig. S2). The lowest methanogenesis efficiency was observed with an oxygen dosage of 50 mg g^{-1} COD_{feed}, which corresponded to the lowest COD removal efficiency (Supplementary Fig. S2), indicating an impaired balance among AD steps (hydrolysis, acidogenesis, acetogenesis and methanogenesis).

The VFA concentrations were lower in the reactors with oxygen dosages of 0, 5, and 10 mg g^{-1} COD_{feed} (90-269, 101-325 and 108-296 mg VFA L^{-1}, respectively), in all stage (Stages 1 to 4) than with higher dosages of 50 and 150 mg g^{-1} COD_{feed} (339-396 and 203-304 mg VFA L^{-1}, respectively) in all stages. The differences in total VFAs between the low-oxygen and high-oxygen groups increased from Stage 1 to Stage 4. The highest VFA accumulation was observed in the reactor with an oxygen input of 50 mg g^{-1} COD_{feed}, corresponding to the lowest methanogenesis efficiency.

The methanogenesis efficiency and VFA accumulation differences indicate a segregation of microbial communities between low micro-aeration dosages (0, 5, and 10 mg g^{-1} COD_{feed}) and

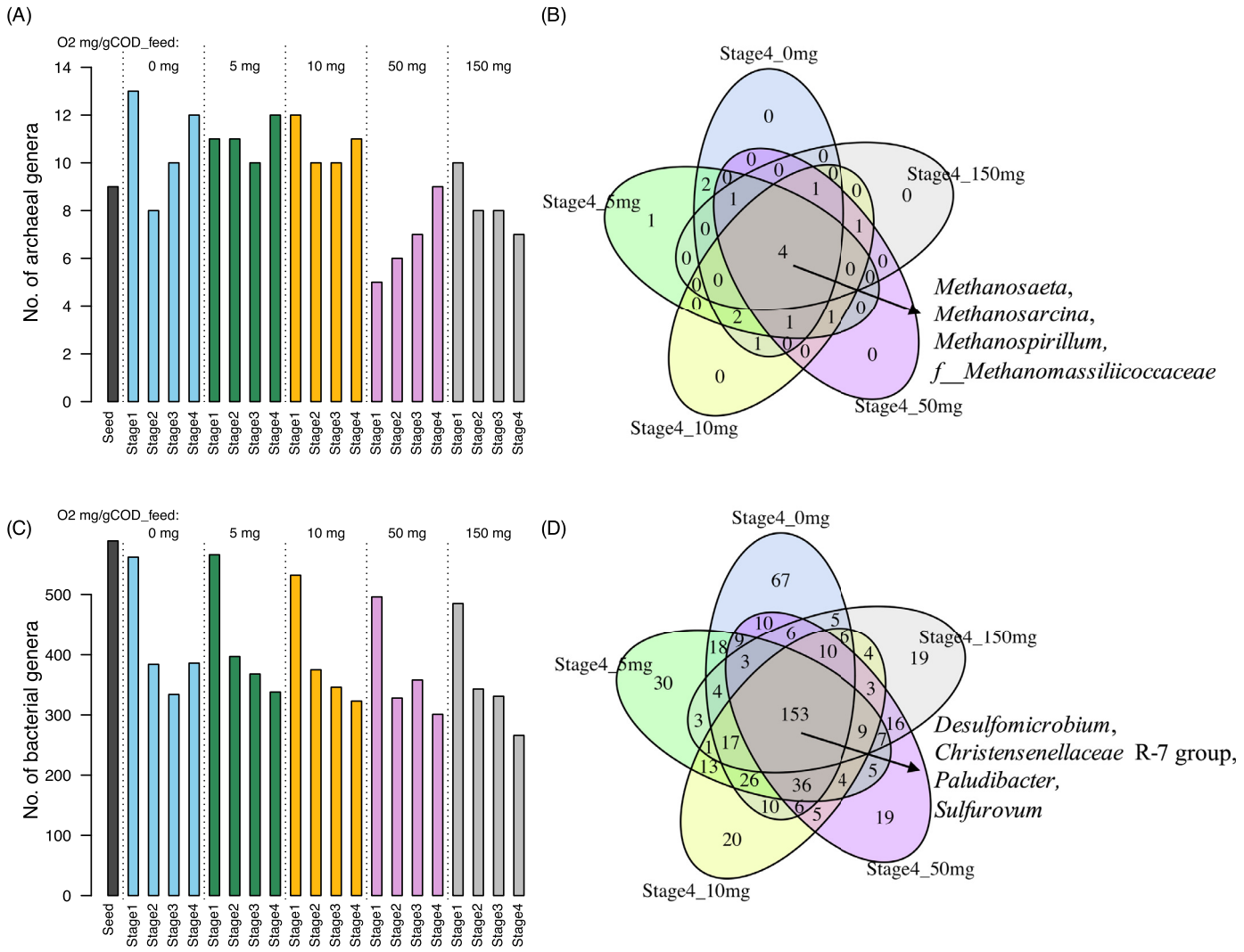


Fig. 2. (A) Archaeal community richness represented by number of genera. (B) Number of shared and unique archaeal genera in Stage 4 (HRT 2d). (C) Bacterial community richness represented by number of genera. (D) Number of shared and unique bacterial genera in Stage 4.

high oxygen dosages (50 and 150 mg g^{-1} COD_{feed}). The oxygen inhibition on methanogenesis was initiated when oxygen dosage exceeded 5 mg g^{-1} COD_{feed}, while the acidogenesis process was not impeded. When the oxygen dosage increased to 150 mg g^{-1} COD_{feed}, the acidogenesis process was impaired, indicated by the lower VFA concentrations than those with the 50 mg g^{-1} COD_{feed}. In addition, it is likely that the high oxygen dosage may have caused aerobic consumption of VFAs, as indicated by the higher CO_2 content in the biogas (Supplementary Fig. S3). High levels of VFA accumulation is another inhibitory factor towards methanogenesis. The excessive oxygen at 150 mg oxygen g^{-1} COD_{feed} resulted in VFA oxidation and mitigated VFA accumulation. Thus, reduced VFA accumulation and improved methanogenesis were observed under the 150 mg oxygen g^{-1} COD_{feed} condition, as compared to the 50 mg oxygen g^{-1} COD_{feed} condition.

3.2. Temporal variation of microbial community diversity

Fig. 2A shows the number of observed genera (community richness) in archaeal communities. The archaeal communities in low-range oxygen dosage reactors (0, 5, and 10 mg g^{-1} COD_{feed}) showed generally higher richness (8-13, 10-12, 10-12 genera, respectively) than the communities of high-range oxygen dosage reactors (5-9 and 7-10 genera for 50 and 150 mg g^{-1} COD_{feed} respectively). In Stage 1 (HRT 5 d), the oxygen dosage of 50 mg g^{-1} COD_{feed} greatly reduced the richness to 5 genera, as compared to 13 genera in the anaerobic condition; however, the richness gradually recovered in the following stages. The high oxygen dosage of 150 mg g^{-1} COD_{feed} led to a progressive decrease of community richness from 10 to 7 genera. In Stage 4 (HRT 2 d), a negative correlation (Pearson's correlation coefficient r = -0.96, p = 0.01) was observed between the archaeal community richness and the oxygen dosage (12, 12, 11, 9 and 7 genera for 0, 5, 10, 50 and 150 mg g^{-1} COD_{feed} respectively); the total archaeal relative abundances was also negatively correlated with the oxygen dosage (0.94%, 0.58%, 0.41%, 0.28% and 0.21% for 0, 5, 10, 50 and 150 mg g^{-1} COD_{feed} respectively, Pearson's correlation coefficient r = -0.97, p = 0.005), indicating that higher oxygen dosage may inhibit the growth of several archaea. The Venn diagram in Fig. 2B shows the number of shared and unique archaeal genera in the last stage. Four genera were shared in all reactors, *i.e., Methanosaeta, Methanosarcina, Methanospirillum,* and unidentified genera from the family of *Methanomassiliicoccaceae.*

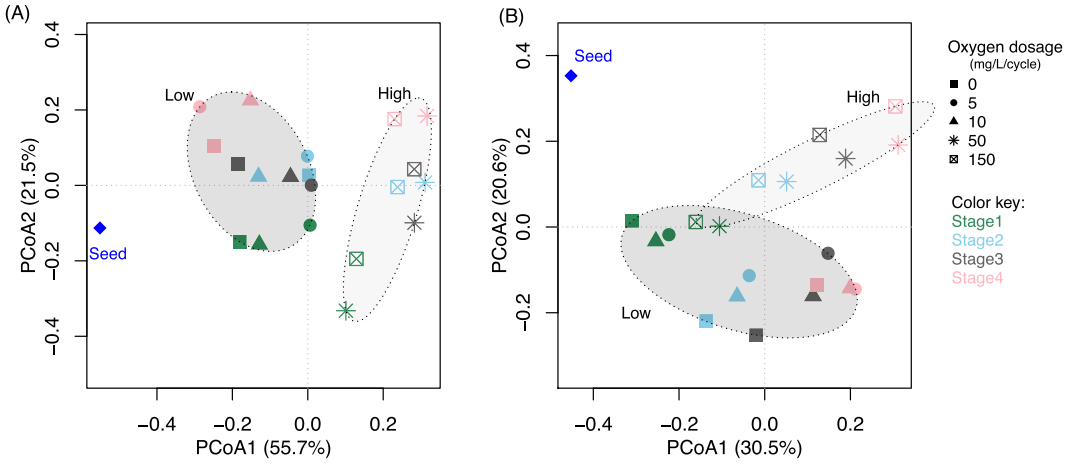


Fig. 3. PCoA of the Bray-Curtis distance of (A) archaeal communities and (B) bacterial communities.

As shown in Fig. 2C, the bacterial community richness exhibited a sharp decrease from Stage 1 to 2 (HRT 5 d to 4 d) for all the 0, 5, 10, 50 and 150 mg g^{-1} COD_{feed} reactors (number of genera from 562 to 384, 566 to 397, 532 to 375, 496 to 328 and 485 to 343, respectively). The number of genera further decreased with reduced HRT to 386, 338, 323, 301 and 266 in the final stage. It is likely that with gradual adaptation and lowering HRT (higher organic loading rate), less diverse but highly functional communities were developed, due to the introduction of higher substrate loading as well as higher concentrations of inhibitory compounds. The high-oxygen-dosage communities (50 and 150 mg g^{-1} COD_{feed}) showed lower bacterial richness, probably because of oxygen inhibition on strictly anaerobic bacteria. There were 153 genera shared in all reactors in the last stage (Fig. 2D), including *Desulfomicrobium, Christensenellaceae* R-7 group, *Paludibacter,* and *Sulfurovum*. Each reactor contained unique genera and the high oxygen dosages led to less unique genera compared to the low oxygen dosages (19, 19, 20, 30, 67 unique genera for 150, 50, 10, 5, 0 mg g^{-1} COD_{feed}, respectively).

The weighted microbial community distances were evaluated using a Bray-Curtis distance matrix and ordinated on a Principal Coordinate Analysis (PCoA) plot (Fig. 3). The archaeal communities (Fig. 3A) showed the highest variation along PCoA1 axis (55.7% of total community variance), associated with the oxygen dosage. A frontier was observed between the low oxygen dosage (0, 5 and 10 mg g^{-1} COD_{feed}) and high oxygen dosage (50 and 150 mg g^{-1} COD_{feed}) conditions. The second axis (PCoA2, 21.5% of total community variance) was associated with the operational stages, differentiating three clusters, Stage 1, Stage 2-3, and Stage 4 communities. Analysis of similarities (ANOSIM) tests indicated significant community differences among oxygen dosages ($r = 0.36$, $p = 0.004$) and among different HRTs ($r = 0.26$, $p = 0.008$). The decrease of HRT corresponds to longer temporal adaptation of microbial communities, leading to higher differences among reactors. Similarly, the bacterial communities (Fig. 3B) also exhibited clusters affected by oxygen dosage and HRT. The temporal variation showed the highest impact, which was presented by the PCoA1 axis explaining 30.5% of community variance. Along the PCoA2 axis (20.6% of community variance), high-oxygen-dosage and low-oxygen-dosage communities were differentiated. ANOSIM tests indicated significantly different bacterial communities among oxygen dosages ($r = 0.18$, $p = 0.031$) and among different HRTs ($r = 0.51$, $p = 0.001$).

For both archaeal and bacterial communities, the distance between high-oxygen-dosage and low-oxygen-dosage communities increased with time. At Stage 4 (HRT of 2 d), the distance between high-oxygen-dosage and low-oxygen-dosage communities was the largest compared to other stages, indicating increased community divergence resulted from the combinatorial effect of oxygen dosage and temporal adaptation. In accordance with the reactor performances (Fig. 1), the microbial communities showed significantly different clusters for high and low-oxygen-dosage conditions for both archaea and bacteria. Our results show that the oxygen dosage was an important niche factor leading to microbial community clusters, possibly based on their different compositions and functions, which is further discussed in the following sections.

3.3. Microbial community composition

3.3.1. Archaeal communities

The relative abundances of archaeal genera based on 16S rRNA gene amplicon sequencing reads are shown in Fig. 4. Different methanogens were enriched at different oxygen dosages. *Methanosaeta* abundances were higher in the three low-oxygen-dosage communities (17.0-50.2%) than in the two high-oxygen-dosage communities (1.1-11.4%) in all stages. *Methanobacterium* (4.3-16.6%) and *Methanolinea* (0.6-1.5%) were also enriched in the three low-oxygen-dosage reactors in the final stage but not detected in the high-oxygen-dosage reactors. *Methanosarcina* were not detected in inoculum but increased during the acclimatization and grew to the third highest methanogen in the last stage (11.4%) in the 5 mg g^{-1} COD_{feed} oxygen dosage reactor. *Methanospirillum* and one unclassified genus in the family *Methanomassiliicoccaceae* were higher in the high-oxygen-dosage reactors (25.7-26.3% and 35.0-41.1% respectively) than in the low-oxygen-dosage reactors (7.4-8.0% and 17.7-23.4% respectively) in the final stage. The responses of methanogens indicate that *Methanosaeta* and *Methanobacterium* are sensitive to oxygen, while *Methanospirillum* and the unclassified genus in the family *Methanomassiliicoccaceae* can tolerant different ranges of oxygen dosages.

3.3.2. Bacterial communities

The bacterial community compositions varied in different reactors at different stages. At the phylum level (Fig. 5), *Bacteroidetes* was the most prevalent phylum in the low-oxygen-dosage reactors at 0, 5, 10 mg g^{-1} COD_{feed} (45.2 ± 6.7 %, 44.4 ± 3.8%, 44.9 ±3.4%, respectively), followed by *Proteobacteria* (15.6 ± 3.2%, 15.7 ± 4.2%, 15.9 ± 4.1%, respectively), and *Firmicutes* (13.9 ± 1.2%, 16.0 ± 3.6%, 15.3 ± 11.4%, respectively). *Bacteroidetes* and *Firmicutes* are known to include functional hydrolysers and fermenters in anaerobic digestion reactors and can be promoted by low-oxygen range micro-aeration (Fu et al., 2016). *Bacteroidetes* and *Firmicutes* de-

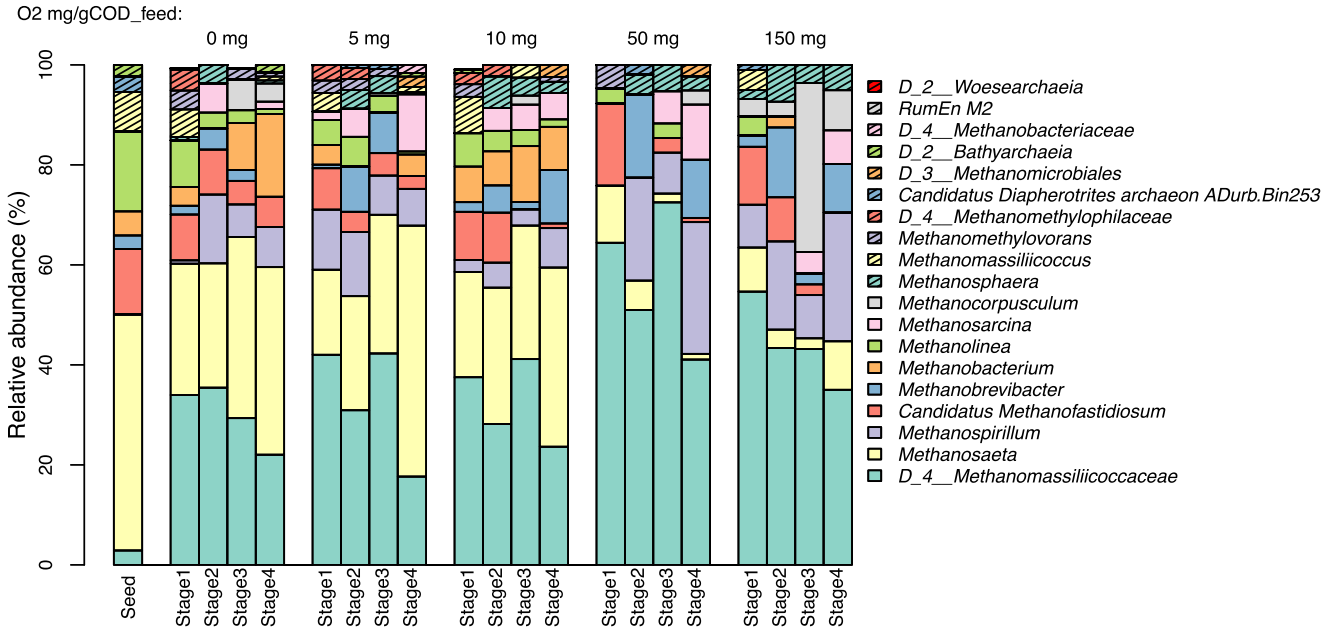


Fig. 4. Relative abundances of archaeal genera (>1% in any sample). Taxon names are shown at genus level, or higher level if not identified, *i.e.*, family level (D_4), order level (D_3), and class level (D_2).

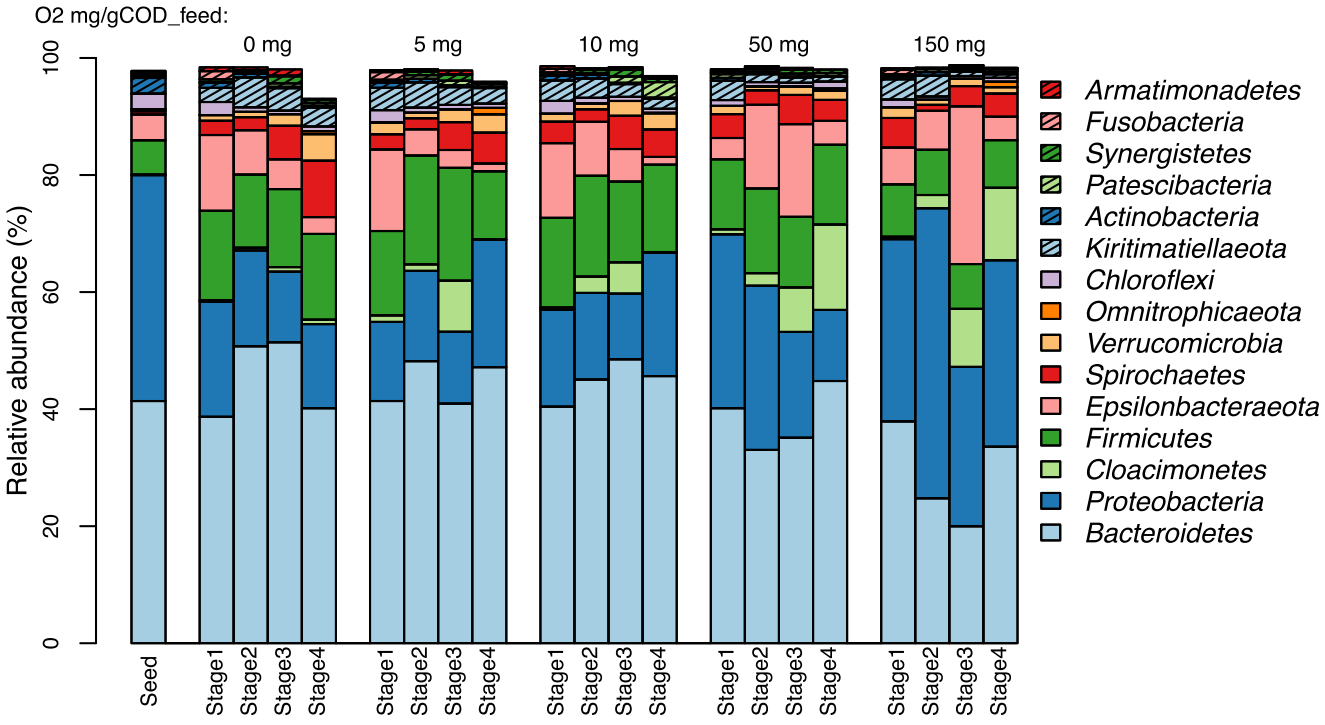


Fig. 5. Bacterial communities at the phylum level (>1% in any sample).

creased with the increased oxygen dosage at 50 mg g^{-1} COD_{feed} (38.3 ± 5.3% and 13.0 ± 1.2%, respectively), and further decreased at 150 mg g^{-1} COD_{feed} (29.1 ± 8.2% and 8.1 ± 0.6%, respectively). *Proteobacteria* increased with a higher oxygen dosage (22.0 ± 8.3% and 34.9 ± 9.9% for 50 and 150 mg g^{-1} COD_{feed}, respectively).

At the genus level, all reactors shared some abundant microorganisms (Supplementary Fig. S4), including *Sulfurovum, Desulfomicrobium, Christensenellaceae* R-7 group, and *Paludibacter*. The high-oxygen-dosage communities (50 and 150 mg g^{-1} COD_{feed}) showed higher abundances than the low-oxygen-dosage communities (0, 5 and 10 mg g^{-1} COD_{feed}) for *Zoogloea, Dechloromonas, Geobacter*, and *Lentimicrobium*, indicating their possible enrichment in microaerophilic and acidogenic conditions. The low-oxygen-dosage reactors were higher in relative abundances of certain bacteria, including *Anaerocella, Macellibacteroides*, and *Treponema*, inferring their low tolerance to oxygen.

3.4. Differential marker microorganisms

Statistical analyses were performed between high and low-oxygen-dosage microbial niches to determine variance amongst microorganisms (markers) using LEfSe (Linear Discriminant Anal-

ysis Effect Size) (Segata et al., 2011). Within the high and low-oxygen-dosage samples, 122 taxa were present at significantly different levels. The phylogenetic tree of the marker microorganisms is shown for phylum to order levels (Fig. 6A) and at the genus level (Fig. 6B).

3.4.1. Enrichment under low oxygen dosage

At the phylum/class/order levels (Fig. 6A), methanogens (*i.e., Methanobacteriales* and *Methanosarcinales*) were enriched within the microbial communities in low-oxygen-dosage environments, which included *Methanobacterium, Methanolinea*, and *Methanosaeta* at the genus level (Fig. 6B). Bacterial phyla, *Armatimonadetes, Bacteroidetes, Deferribacteres, Fibrobacteres*, WPS_2, *Verrucomicrobia*, and *Tenericutes*, were all significantly higher under the low-range oxygen dosages (Fig. 6A), indicating inhibition effect on them at higher oxygen dosages.

The enriched bacteria included syntrophic VFA degraders, *e.g., Syntrophobacteriales* are known as syntrophic propionate fermenters (Zhang et al., 2009), *Syntrophomonas* degrade butyrate (Muller et al., 2010). The syntrophic partnership between these bacteria and methanogens may explain the lower levels of VFA accumulation and higher methanogenesis at low-oxygen dosages than high-oxygen dosages (Fig. 1). The enhanced methanogenesis and VFA utilization provide positive feedback to the hydrolysis; whereas high VFA concentrations can inhibit the hydrolysis step. Some bacteria have been reported to ferment under anaerobic conditions and degrade long-chain fatty acids, e.g., *Microthrix* (McIlroy et al., 2013), *Cytophaga* (Anderson and Ordal, 1961; Bachmann, 1955; Starns et al., 2014), SJA_88 (Hatamoto et al., 2007), *Desulfomicrobium* (Copeland et al., 2009). Several genera can survive under anaerobic meso-aerophilic environments, *e.g., Macellibacteroides* (Jabari et al., 2012), *Petrimonas* (Grabowski et al., 2005), *Proteocatella* (Pikuta et al., 2009), *Dehalobacterium* (Chen et al., 2017), *Acidaminobacter* (Stams and Hansen, 1984), *Anaerovorax* (Matthies et al., 2000), *Sedimentibacter* (Imachi et al., 2016), and *Victivallis* (Zoetendal et al., 2003).

In the reactor of 5 mg g^{-1} COD_{feed}, two *Bacteroidetes* genera, M2PB4-65 termite group and vadinHA17, and *Desulfomicrobium*, were higher than in other reactors (Supplementary Fig. S4). *Bacteroidetes* are important hydrolytic and acidogenic bacteria (Fu et al., 2016) and *Desulfomicrobium* can utilize propionate to produce acetate (Xiong et al., 2017), which may have contributed to the higher hydrolysis and methanogenesis efficiencies at 5 mg g^{-1} COD_{feed} (Fig. 1).

3.4.2. Enrichment under high oxygen dosages

The high-oxygen-dosage environments led to significantly higher abundances of the phyla *Chlorobi* and WWE1, including the orders *Ignavibacteriales* and *Cloacamonales*, respectively (Fig. 6A). *Ignavibacteriales* contain facultative anaerobic species with functions of producing glycoside hydrolases for plant biomass deconstruction (Hiras et al., 2016) and cellulolytic activity (Liu et al., 2012). *Cloacamonales* (phylum WWE1) have been reported to be enriched in cellulose degradation processes in anaerobic digestion (Limam et al., 2014). Even though these hydrolytic bacteria are enriched under high-oxygen dosages, their activity may be negatively affected by the accumulated VFAs since there is a lack of syntrophic VFA degradation and methanogenesis.

The enriched bacteria showed resistance to oxygen in anaerobic environment. *Elusimicrobiales* contains a single species, *Elusimicrobium minutum*, which is a facultative anaerobic ultramicrobacterium and has an oxygen stress protection gene cluster to resist oxygen (Herlemann et al., 2009). Genera *Zoogloea* and *Dechloromonas* (order *Rhodocyclales*) are commonly reported in anaerobic-aerobic wastewater treatment systems operated for biological nutrient removal. *Zoogloea* are aerobic and *Dechloromonas* are facultative-anaerobic chemoorganoheterotrophic organisms that are rapidly selected in the presence of organic matter and a terminal electron acceptor like dissolved oxygen (Saunders et al., 2016; Weissbrodt et al., 2012; Zhang et al., 2012). *Thiotrichales* (genus *Thiothrix*, Fig. 6B) is aerobic or microaerophilic (Garrity et al., 2005). The genus *Clostridium* can resist microoxic conditions by consuming oxygen then resume growth without oxygen, with VFA production functions, *e.g., C. acetobutylicum* (Kawasaki et al., 2005) and *C. butyricum* (Kawasaki et al., 1998). The sulfate-reducing bacteria *Desulfovibrio, Desulfobacter*, and *Desulforhopalus* can tolerate low-level oxygen (Cypionka et al., 1985; Fournier et al., 2003; van Niel and Gottschal, 1998). The sulfur-oxidizing bacteria, *Sulfuricurvum*, were also enriched in this study, and were reported to be facultative-anaerobic (Kodama and Watanabe, 2004).

3.5. Relative abundances of MGEs and ARGs

As shown in Fig. 7, the mobile gene elements (MGEs) and ARGs were higher in the low-oxygen-dosage conditions than in the high-oxygen-dosage conditions. Two-way ANOVA test (HRT and oxygen dosage) showed that all the ARGs were significantly correlated with the oxygen dosage applied ($p < 0.05$), but not HRT nor the interacted effect of HRT and oxygen ($p > 0.05$) (Supplementary Table S2), implying that the micro-aeration dosages play a significant role in controlling the abundances of MGEs and ARGs.

Bacteria can transfer resistance genes using MGEs such as integrons and plasmids, which were quantified using the class 1 integrase gene *intI1* (Ploy et al., 2000) and IncP-1 plasmid RK2 specific gene *korB* (Kornacki et al., 1987; Pallares-Vega et al., 2019). The relative abundance of *korB* was higher in all the reactors than blackwater, while *intI1* increased in low oxygen communities but not in high-oxygen-dosage communities (50 and 150 mg g^{-1} COD_{feed}). Since both MGEs showed relatively low abundances in the feeding blackwater, their increases in the reactors were likely induced by the oxygen dosages and enriched microbial community rather than from the feeding. Oxygen level has been found to affect the MGEs in other studies. A study comparing aerobic activated sludge and anaerobic digestion sludge showed that AD had higher integrons and conjugative transposons but lower plasmids than the aerobic sludge (Osińska, 2019). To the contrary, another study found that integron and plasmids were much higher in the aerobic sludge than in the AD sludge (Tao et al., 2016).

As shown in Fig. 7, the *tetM* was relatively low in blackwater and inoculum, while it increased in low-oxygen-dosage communities and remained low in high-oxygen-dosage communities. The *ermB* was relatively higher than *tetM* in blackwater and inoculum and varied temporally. In the final stage, the 150 mg g^{-1} COD_{feed} dosage reactor showed the lowest *tetM* and *ermB. Similarly, sul1* and *sul2* were relatively higher in the low-oxygen-dosage communities than in high-oxygen-dosage communities, while *blaCTX-M* and *qnrS* were not abundant in blackwater, inoculum, or SBRs.

The Mantel test was performed to examine the correlation (Pearson's product-moment correlation, $r = 0.34$) between microbial community and ARGs' abundances, with p-level < 0.01. Canonical Correspondence Analysis (CCA) also shows that ARG levels are correlated with low oxygen dosage communities (Supplementary Fig. S5, $p < 0.01$), indicating that the ARG profiles were affected largely by the indigenous microbial community. The microbial communities host more MGEs (*intI1* and *korB*) in low-oxygen conditions than in high-oxygen micro-aerobic processes, which may lead to similar trends of the prevalence of the measured ARGs. Our study was not able to identify which taxa contributed to the MGEs and future studies using metagenomic approaches are suggested to relate the MGEs and specific species contributions. Du et al. (2015) reported that anaerobic and anoxic

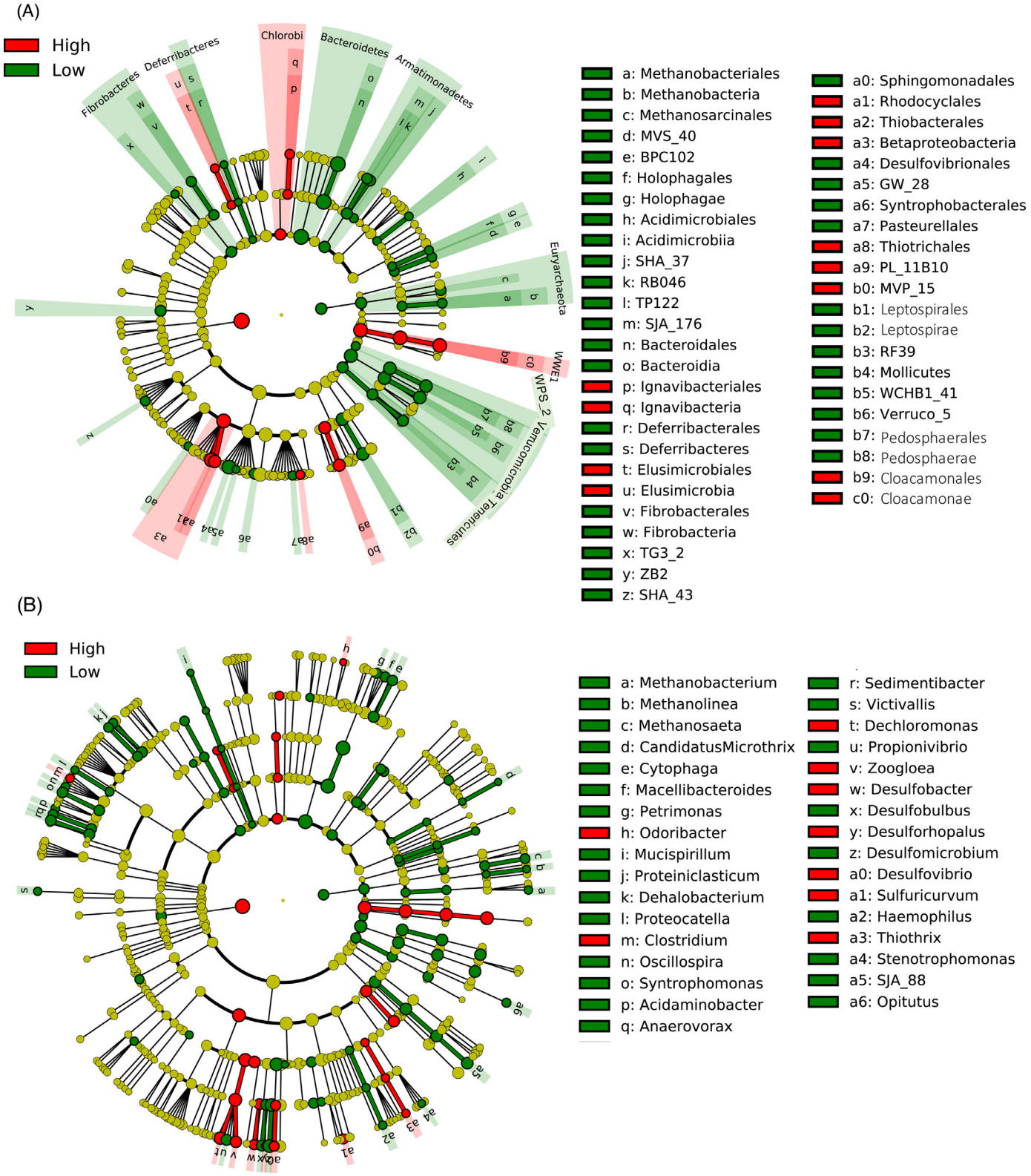


Fig. 6. Linear discriminant analysis Effect Size (LEfSe) analysis comparing microbial niches established under high and low oxygen dosages in the blackwater AD systems for Stage 2-4. (A) Taxa shown at phylum, class, and order levels. (B) Taxa shown at genus level. Red color of points and shades indicates significant higher abundances in high-oxygen dosage conditions (50 and 150 mg g^{-1} COD_{feed}), green color indicates significant higher abundances in low-oxygen dosage conditions (0, 5 and 10 mg g^{-1} COD_{feed}).

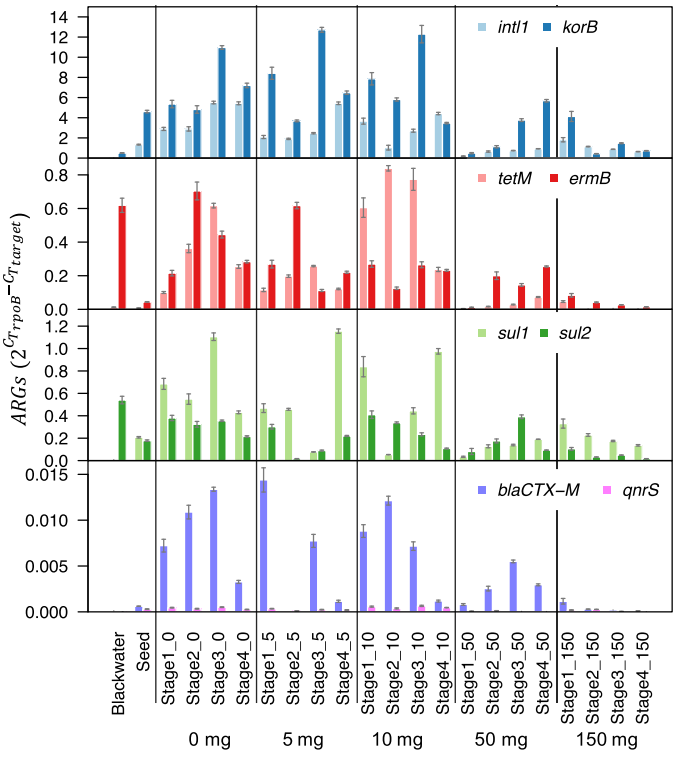


Fig. 7. Relative quantification of MGEs (class 1 integrase gene *intI1* and IncP-1 plasmid RK2 specific gene *korB*) and ARGs tetracyclines (*tetM*), macrolides (*ermB*), sulfonamides (*sul1* and *sul2*), extended-spectrum beta-lactamases (*blaCTX-M*), and fluoroquinolones (*qnrS*) by qPCR with reference to bacteria *rpoB* gene ($2^{C_{T\,rpoB}-C_{T\,target}}$). Genes at similar relative levels were put in the same panels. Error bar represents three replicates in qPCR test.

basins perform better than aerobic basin in removing ARGs (*sul1, intI, tetX, tetG, tetW*) in the anaerobic/anoxic/aerobic wastewater treatment process, while the ratio of ARGs to 16S rRNA gene in the sludge remained the same. It was hypothesized that microorganisms have lower bioactivity under anaerobic condition compared to aerobic condition thus limiting the propagation of ARGs (Du et al., 2015; Pei et al., 2007). However, agreeing to our results, some studies showed that AD sludge host more ARGs than aerobic sludge. Yang et al. (2014) showed that AD sludge carried more ARGs than aerobic activated sludge using metagenomic sequencing. The average abundance of ARGs in activated sludge (0.2 copy of ARGs per cell) was less than AD sludge (0.43 copy of ARGs per cell) (Yin et al., 2019). It has been reported that aerobic processes had better ARG reduction than anaerobic processes in wastewater treatment systems (Tao et al., 2014; Yuan et al., 2016).

The mechanisms leading to more ARGs in AD than in aerobic sludge include a few hypotheses. Oxygen concentrations decrease towards the biofilm interior, and the lower oxygen and lower metabolic activity were correlated with higher antibiotic tolerance (Walters et al., 2003), indicating that lower metabolic activity in anaerobic conditions may result in higher relative ARG levels compared to aerobic conditions. Another theory is that aerobic sludge microorganisms host higher proportions of efflux pump and target modification genes than anaerobic sludge as revealed by metagenomic analysis (Christgen et al., 2015), thus aerobic and anaerobic-aerobic sequence reactors were more effective in reducing ARGs than the anaerobic reactors. We compared the predicted efflux pump genes resulted from PICRUSt2 (Douglas et al., 2020) analysis of the 16S rRNA gene sequences using KEGG antimicrobial resistance gene database (Kanehisa, 2018) (Supplementary Table S3), which revealed that a few genes were higher in the high-oxygen dosage conditions, including efflux pumps AdeABC (K18143, K18144, K18145, K18146), AbcA (K18104), MexJK-OprM (K18303), BpeEF-OprC (K18900, K18901, K18902, K18903) and MexPQ-OpmE (K19591) (Supplementary Fig. S6). Metagenomic analysis is suggested to elucidate the true efflux pump gene abundances.

Using isolated *Escherichia coli*, it was remonstrated that semi-aerobic condition can reduce gene transfer by plasmids between E. coli in landfill leachate, compared to anaerobic condition (Threedeach et al., 2012). A recent study reported the impact of redox conditions on conjugative gene transfer frequency in wastewater treatment processes, where aerobic conditions showed lower gene transfer than anaerobic conditions (Jong et al., 2020). It demonstrated that protozoa predation under aerobic conditions contributed to plasmid reduction, but other mechanisms of the oxygen stress on ARG transfer remains unclear. Our results showed that the gene transfer by plasmids and class 1 integrons were less when oxygen was supplied at 50-150 mg g^{-1} COD_{feed} than at 0-10 mg g^{-1} COD_{feed}, leading to reduced ARGs in the reactor. Moreover, different microorganisms vary in abilities of horizontal gene transfer and antibiotic resistance (Yang et al., 2014; Yin et al., 2019), thus microbial community composition is a key factor in determining relative ARG levels.

Combining the reactor performance results and the prevalence of MGEs and ARGs, the low-oxygen dosage conditions showed higher hydrolysis and methanogenesis efficiencies (Fig. 1), but also higher levels of MGEs and ARGs (Fig. 7) than the high-oxygen dosage conditions, suggesting that operational strategies using aerobic or high-dosage micro-aerobic processes can help to combat ARGs in wastewater treatment processes. Applying aerobic or high-dosage micro-aerobic conditions before discharging biosolids to the environment may help to reduce ARG transfer. Another strategy is to use high-dosage micro-aeration in AD process for harvesting VFAs meanwhile reducing ARGs.

4. Conclusions

Anaerobic digestion of blackwater at ambient room temperature can be improved by micro-aeration at proper dosages. Different oxygen dosages (0, 5, 10, 50, and 150 mg g^{-1} COD_{feed}) were applied to anaerobic sequencing batch reactors. Varying micro-aeration dosages resulted in significant differences in archaeal and bacterial communities ($p < 0.05$). The community dissimilarity (beta-diversity) and reactor performance differences indicated the formation of different microbial ecological niches namely a low-oxygen-dosage niche (0, 5, and 10 mg g^{-1} COD_{feed}) and a high-oxygen-dosage niche (50 and 150 mg g^{-1} COD_{feed}). The low-oxygen-dosage niche (0-10 mg g^{-1} COD_{feed}) functioned well for hydrolysis and methanogenesis while high-oxygen-dosage niche (50-150 mg g^{-1} COD_{feed}) led to VFA accumulation. Statistical analysis using linear discriminant analysis effect size (LEfSe) revealed the differential marker microorganisms in these two niches. Anaerobic syntrophic bacteria (*e.g., Syntrophomonas*) and certain methanogens (*Methanobacterium, Methanolinea*, and *Methanosaeta*) were enriched in the low-oxygen-dosage niche. Facultative anaerobic bacteria and aerobic bacteria were markers of the high-oxygen dosage-niche (*e.g., Ignavibacteriales, Cloacamonales, Rhodocyclales*). Since blackwater may contain substances that stimulate ARGs, the ARGs and related MGEs in blackwater and in the reactors were quantified relatively with reference to total bacteria. The prevalence of the measured ARGs (*tetM, ermB, sul1, sul2, blaCTX-M*) were higher in the low-oxygen-dosage niche samples possibly due to the higher MGEs (*intI1* and *korB*) and distinct microbial community compositions. The application of micro-aeration for blackwater treatment is greatly affected by the oxygen dosage, as it determines the microbial community niche as well as ARG prevalence. Collectively, this study highlights that appropriate oxygen input to

AD systems can help to manage the reactor performance and ARG levels, which provides a strategy for future AD applications.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

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